

Background



- The UWAFIT team wants to mount two LiDAR sensors on the front and back
- This would enable CAV to use the LiDAR for low-speed parking scenarios
- UWAFIT goals is to implement these two LiDAR sensors with keeping functionality and aesthetics in mind

Design Constraints

1) Mount Angle $\leq 10^\circ$

2) Field of View $\geq 120^\circ$

3) Modular Design

4) Must fit in 3" Envelope

Design Criteria

1) Weatherproofing

2) Aesthetics

3) Structural Rigidity

4) Cost Effectiveness

Task ID	Task Title	Start Date	Due Date	Pct of Task Complete	September				October				November				December			
					W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
1	LiDAR Placement		2025-10-06																	
1.1	Positional Research / Background Research	09-22-25	09-26-25	100%																
1.2	Define Design Requirements	09-22-25	09-26-25	100%																
1.3	Location Mapping / Prelim CAD Dimensions	10-01-25	10-06-25	100%																
2.0	Prototyping																			
2.1	Design																			
2.2	Software Learning (CAD, Siemens NX)	09-25-25	10-28-25	100%																
2.3	Mount Design - Front	10-06-25	10-28-25	100%																
2.4	Mount Design - Rear	10-06-25	10-28-25	100%																
2.7	Simulations																			
2.8	Vibrations	10-29-25	11-03-25	100%																
2.9	FEA	10-30-25	11-04-25	100%																
2.11	Vehicle Setup For Mounting																			
2.12	Protoyping	11-05-25	11-12-25	100%																
2.13	Integration																			
2.14	Electrical Routing	11-05-25	11-12-25	0%																
3	Aestheics																			
3.1	Material Selection	10-27-25	11-03-25	100%																
3.2	Surface Modeling	11-03-25	11-10-25	100%																
3.3	Manufacturing Methods	11-10-25	11-17-25	100%																
4.0	Testing and Calibration																			
4.1	LiDAR Calibration/ Validation	11-17-25	11-24-25	0%																